

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : Civil

Semester :V

Subject Code & Name: Geotechnical Engineering (BTCVC502)

Max Marks: 60

Date : 31/01/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Explain two phase and three phase system of soil with a neat sketch.	L2 /1	6
B) Define moisture content. Enlist various method of determination of moisture content and explain anyone method in detail.	L2 /1	6
C) Explain soil structure with neat sketches.	L2 /1	6
Q.2 Solve Any Two of the following.		12
A) Define Specific Gravity. Enlist various method of determination of Specific Gravity and explain anyone method in detail.	L2 /1	6
B) Derive an expression between bulk density, water content and dry density.	L3 /1	6
C) A partially saturated soil has water content of 19% and bulk unit weight of 20KN/m³. Assume G=2.6 calculate degree of saturation, void ratio and porosity.	L3 /1	6
Q. 3 Solve Any Two of the following.		12
A) List various methods of determination of coefficient of permeability and explain anyone method with a neat sketch.	L2 /2	6
B) A falling head permeability test was conducted on a sample of diameter 6cm and height 15cm. Diameter of stand pipe was 2cm, initial head=45cm and final head=30cm. Time elapsed =1min 45sec, determine K.	L3 /2	6
C) What are the different factors affecting permeability of soil.	L2 /2	6

Q.4	Solve Any Two of the following.		12
A)	Explain different types of shear test conditions of conduction.	L2 /3	6
B)	Explain the procedure to conduct vane shear test on a soil specimen with a neat sketch.	L2 /3	6
C)	A cylindrical sample of soil having cohesion of 0.8Kg/cm^2 and angle of internal friction of 20°, is subjected to a cell pressure of 1.0Kg/cm^2. Calculate maximum deviator stress at which the sample will fail and the angle made by failure plane with the axis of sample.	L3 /3	6
Q. 5	Solve Any Two of the following.		12
A)	Explain spring analogy method of consolidation process.	L2 /3	6
B)	A soil sample has OMC of 10% and bulk density of 1.80gm/cc. Determine void ratio, degree of saturation and dry density take $G=2.7$	L3 /3	6
C)	Explain the difference between Standard Proctor test and Modified Proctor test.	L4 /3	6

***** End *****